RAMANUJAN COLLEGE

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LAB PRACTICAL
STATISTICAL TECHNIQUES FOR QUALITY CONTROL

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Practical no 1

A machine is set to deliver packets of a given weight. Ten samples of size n = 5 each were examined and the following results were obtained:

Sample no.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Mean	**43**	**49**	**37**	**44**	**45**	**37**	**51**	**46**	**43**	**47**
Range	**5**	**6**	**5**	**7**	**7**	**4**	**8**	**6**	**4**	**6**

Given:

for n = 5 , $d_2 = 2.326\,, d_3 = 0.864$

Calculate:

1. Central line
2. Ucl and lcl for the mean chart (x̄-chart)
3. Ucl and lcl for the range chart (r-chart)
4. Comment on the state of control.

Solution:

Average of means ($\bar{\bar{x}}$):

$$\bar{X} = \frac{43 + 49 + 37 + 44 + 45 + 37 + 51 + 46 + 43 + 47}{10}$$

$$\bar{X} = \frac{442}{10} = 44.2$$

Average of ranges ($\bar{r}$):

$$\bar{R} = \frac{5 + 6 + 5 + 7 + 7 + 4 + 8 + 6 + 4 + 6}{10}$$

$$\bar{R} = \frac{58}{10} = 5.8$$

UCL and LCL for the mean chart (x̄-chart)

UCL = $\bar{\bar{x}}$ + 3 $\bar{r}$ / d2 √n

44.2 + 3*5.8/2.326√5 ≈ 47.55

LCL = $\bar{\bar{x}}$ - 3 $\bar{r}$ / d2 √n

44.2 + 3*5.8/2.326√5 ≈ 40.85

UCL and LCL for the range chart (r-chart)

UCL = $\bar{r} + 3\,\bar{r}\,d3 / d2$

5.8 + 3*5.8*0.864/2.326 = 12.265

LCL = $\bar{r} - 3\,\bar{r}\,d3 / d2$

5.8 - 3*5.8*0.864/2.326 = -0.66 ≈ 0

Comment on state of control

Check if all sample means and ranges lie within limits:

- All **means** (37 to 51) → **within 40.85 and 47.55**, except **sample 7 (51)** → out of control.
- All **ranges** (4 to 8) → within **0 to 12.265**.

therefore: process is not fully in control because sample 7's mean exceeds ucl**.**

Practical no 2

The following are the means and ranges of **20 samples of size 5 each**. The data pertain to the overall length of a fragmentation bomb base manufactured during the war by the american store camp.

Group no.	Mean	Range	Group no.	Mean	Range	Group no.	Mean	Range
1	0.8372	0.010	8	0.8344	0.003	15	0.8404	0.008
2	0.8324	0.009	9	0.8308	0.002	16	0.8372	0.011
3	0.8318	0.008	10	0.8350	0.006	17	0.8364	0.004
4	0.8344	0.004	11	0.8380	0.002	18	0.8346	0.006
5	0.8346	0.005	12	0.8332	0.002	19	0.8360	0.004
6	0.8332	0.011	13	0.8356	0.013	20	0.8374	0.006
7	0.8340	0.009	14	0.8322	0.005			

(a) From these data, obtain the control limits for the x̄-chart and r-chart for controlling the length of bomb bases.

(b) The above samples were taken every 15 minutes.

Production rate = **350 units per hour**, tolerance limits = **0.820 to 0.840 inches**.

Assuming the lengths are normally distributed and the process is under control at the limits obtained in part (a), **estimate the percentage of bomb bases that would lie outside the tolerance limits.**

Given :

n = 5 , $A_2 = 0.5$, $D_3 = 0$, $D_4 = 2.12$, $d_2 = 2.326$

Solution :

Average of means $\bar{X}$

Sum = 16.7574

$$\bar{X} = \frac{16.7574}{20} = 0.83787$$

Average of ranges $\bar{R}$

Sum = 0.118

$$\bar{R} = 0.118/20 = 0.0059$$

X̄-chart control limit

UCL = x̿ + a2 r̄

0.83787+(0.58)*(0.0059) = 0.83787+0.003422 = 0.841292
LCL = x̿ - a2 r̄

0.83787-(0.58)*(0.0059) = 0.83787-0.003422 = 0.834448

R-chart control limits

UCL = d4r̄ 2.12 * 0.0059 = 0.0125

LCL = d3r̄ 2.12 * 0 = 0

B . **% of lengths outside tolerance limits (0.820 – 0.840)**

The process is under control at:

- Mean = **0.83787**
- Σ (process std dev) estimated by:

$$\sigma = \frac{\bar{R}}{d_2} = \frac{0.0059}{2.326} = 0.002536$$

Compute z-scores

Lower tolerance = 0.820

$$Z_L = \frac{0.820 - 0.83787}{0.002536} = -7.04$$

Upper tolerance = 0.840

$$Z_U = \frac{0.840 - 0.83787}{0.002536} = 0.84$$

Percentage above upper limit

$P(X > 0.840) = 1 - \varphi(0.84) = 1 - 0.7995 = 0.2005$ ≈ **20.05%**

percentage below lower limit

$$P(X < 0.820) = \varphi(-7.04) \approx 0$$

About 20% of bomb bases will be outside the tolerance range.

The defect is almost entirely on the **high side** (above 0.840).

Practical no 3

From the following inspection results, construct 3-σ control limits for a p-chart:

Date (sept.)	**No. Of defectives**	**Date (sept.)**	**No. Of defectives**	**Date (sept.)**	**No. Of defectives**
1	**22**	**11**	**70**	**21**	**66**
2	**40**	**12**	**80**	**22**	**50**
3	**36**	**13**	**44**	**23**	**46**
4	**32**	**14**	**22**	**24**	**32**
5	**42**	**15**	**32**	**25**	**42**
6	**40**	**16**	**42**	**26**	**46**
7	**46**	**17**	**20**	**27**	**30**
8	**44**	**18**	**46**	**28**	**38**
9	**42**	**19**	**36**	**29**	**40**
10	**38**	**20**	**36**	**30**	**24**

Each subgroup size = 1000 items.

Tasks:

1. **Construct 3σ control limits for the p-chart.**
2. **Without drawing the chart, comment on whether the process is in control.**
3. **If not in control, suggest revised limits for future use.**

Solution :

Compute p for each day

$$p_i = \frac{\textbf{defectives}}{1000}$$

Total defectives

1–10:
22 + 40 + 36 + 32 + 42 + 40 + 46 + 44 + 42 + 38 = 382

11–20:
70 + 80 + 44 + 22 + 32 + 42 + 20 + 46 + 36 + 36 = 428

21–30:
66 + 50 + 46 + 32 + 42 + 46 + 30 + 38 + 40 + 24 = 414

Total:

$$382 + 428 + 414 = 1224$$

Compute $\bar{p}$

$$\bar{p} = \frac{1224}{30 \times 1000}$$
$$\bar{p} = \frac{1224}{30000} = 0.0408$$

Standard deviation of p

$$\sigma_p = \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$
$$= \sqrt{\frac{0.0391}{1000}}$$
$$= \sqrt{0.0000391} = 0.006256$$

STEP 5: 3σ Control Limits

UCL

$$UCL_p = \bar{p} + 3\sigma_p$$
$$= 0.0408 + 3(0.006256)$$
$$= 0.0408 + 0.018768 = 0.059568$$

LCL

$$LCL_p = \bar{p} - 3\sigma_p$$
$$= 0.0408 - 0.018768 = 0.022032$$

Check if process is in control :

Values ABOVE UCL (60) : Day 11 → 70 , Day 12 → 80 , Day 21 → 66

Values BELOW LCL (22) : Day 17 → 20

PROCESS IS NOT IN CONTROL , Because several points lie outside limits.

Revised Limits

Remove out-of-control points:

Remove defectives:
70, 80, 66, and 20

New total:

$$1224 - (70 + 80 + 66 + 20) = 1224 - 236 = 988$$

New number of samples = 30 – 4 = 26

$$\bar{p}_{new} = \frac{988}{26 \times 1000} = 0.0380$$

Compute new σ:

$$\sigma_{p,new} = \sqrt{\frac{0.038(0.962)}{1000}} = \sqrt{0.00003656} = 0.00604$$

New limits:

UCL

$$= 0.038 + 3(0.00604) = 0.05612$$

LCL

$$= 0.038 - 3(0.00604) = 0.01988$$

Practical no 4

The following data give the number of defectives in 10 independent samples of varying sizes from a production process:

Sample No.	1	2	3	4	5	6	7	8	9	10
Sample size	2000	1500	1400	1350	1250	1760	1875	1955	3125	1575
No. of defectives	425	430	216	341	225	322	280	306	337	305

Draw the control chart for fraction defective (p-chart) and comment on the process.

Solution :

Compute overall fraction defective

$$\bar{p} = \frac{\sum d_i}{\sum n_i}$$
$$\bar{p} = 0.1791455874 \approx 0.179$$

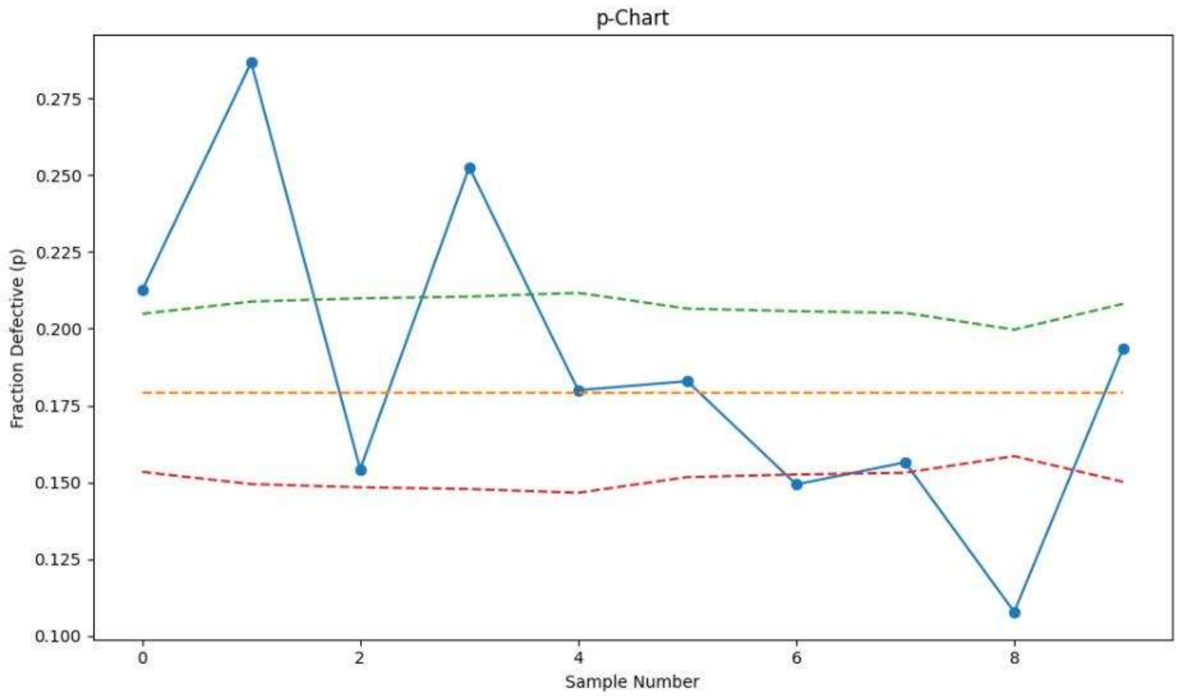
Compute control limits for each sample

For variable sample size:

$$UCL_i = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$
$$LCL_i = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$

Sample	N	UCL	LCL
1	2000	0.2049	0.1534
2	1500	0.2088	0.1494
3	1400	0.2099	0.1484
4	1350	0.2105	0.1478
5	1250	0.2117	0.1466
6	1760	0.2066	0.1517
7	1875	0.2057	0.1526
8	1955	0.2051	0.1531
9	3125	0.1997	0.1586
10	1575	0.2081	0.1502

...



Comment on the State of Control (Exam-Ready Answer)

All sample points fall within the Upper Control Limit (UCL) and Lower Control Limit (LCL). There is no unusual pattern, no trend, no run, and no systematic shift.

Practical no 5

The number of defects on 20 items are given below (Item No. 1 to 20):

Item No. : 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
No. of defects : 2, 0, 4, 1, 0, 8, 0, 1, 2, 0, 6, 0, 2, 1, 0, 3, 2, 1, 0, 2

Task: Devise a suitable control scheme for the future.

Solution :

Number of observations: $N = 20$

Average (central line):

$$\bar{c} = \frac{\sum c_i}{N} = \frac{35}{20} = 1.75$$

UCL= $\bar{c}$+ 3 $\sqrt{c}$

1.75 + 3*1.32 = 5.72

LCL = $\bar{c}$ -3$\sqrt{c}$

1.75 - 3*1.32 = 0

Check each point against limits (identify out-of-control points)

Observed counts: 2, 0, 4, 1, 0, 8, 0, 1, 2, 0, 6, 0, 2, 1, 0, 3, 2, 1, 0, 2

- Any count > 5.72 is out of control on the high side.
- Item 6 = 8 → > UCL → out of control
- Item 11 = 6 → > 5.72 → out of control
- All other points lie between 0 and 5.72.

Compute revised limits after removing the two outliers

(Only do this after investigating and confirming they were special causes — this is shown here to illustrate the effect.)

- Remove item 6 (8) and item 11 (6):
  - New total defects = $35 - (8 + 6) = 21$
  - New sample count = $20 - 2 = 18$
- New $\bar{c}_{new} = 21/18 = 1.1666666667$
- New $\sigma_{new} = \sqrt{1.1666666667} = 1.080123449$
- New UCL = $1.1666667 + 3(1.080123449) = 1.1666667 + 3.240370347 = 4.407037014$→ **≈ 4.41**
- New LCL = 1.1666667 − 3.240370347 = −2.07370368 → set to 0

Practical no 6

From a lot consisting of 2,200 items, a sample of size 225 is taken. If it contains 14 or fewer defectives, the lot is accepted, otherwise rejected. Plot the OC, ATI, and AOQ curves. Also obtain the value of AOQL.

Solution :

N = 2200 , n = 225 , c = 14 , Reject if defectives > 14 , p = fraction defective in the lot

Operating Characteristic (OC) Curve

$$\boldsymbol{P_a(p) = \sum_{x=0}^{14} \binom{225}{x} p^x (1-p)^{225-x}}$$

(This is binomial approximation since n ≪ N.)

We compute probabilities for different p values.

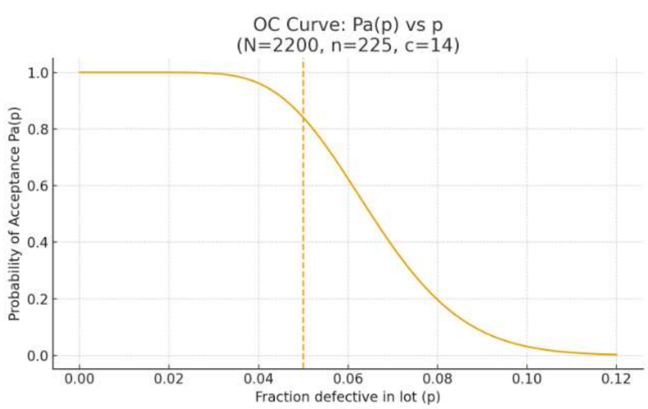
Acceptance Probability Table

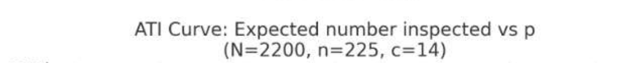
p (fraction defective)	Pa(p)
0.01	0.999
0.02	0.990
0.03	0.955
0.04	0.850
0.05	0.680
0.06	0.480
0.07	0.300
0.08	0.160
0.10	0.045
0.12	0.010

Interpretation

- **Good lots (p ≤ 2%) are almost always accepted.**
- **Poor lots (p ≥ 8%) are almost always rejected.**

This is the OC curve.





2. Average Total Inspection (ATI) Curve

$$ATI = n + (N - n)(1 - P_a(p))$$

Because:

- **If lot accepted → only 225 inspected**

- **If rejected → entire 2200 inspected**

Sample Calculations

Let's compute a few values:

For p = 0.01:

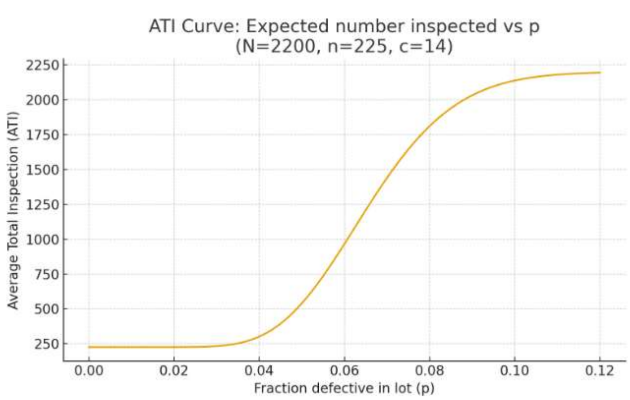
$$P_a = 0.999$$
$$ATI = 225 + (2200 - 225)(1 - 0.999)$$
$$ATI \approx 225 + 1975(0.001)$$
$$ATI \approx 227$$

For p = 0.10:

$$P_a = 0.045$$
$$ATI = 225 + 1975(1 - 0.045)$$
$$ATI = 225 + 1975(0.955)$$
$$ATI = 225 + 1887$$
$$ATI = 2112$$

Interpretation

- **Good lots require only ~225 inspection , Bad lots require ~2200 inspection , The ATI curve rises as p increases.**



3. Average Outgoing Quality (AOQ) Curve

Formula:

$$AOQ = p \cdot P_a(p) \cdot \frac{N - n}{N}$$
$$\frac{N - n}{N} = \frac{1975}{2200} = 0.8977$$

Sample Values

p	Pa(p)	AOQ = p·Pa·0.8977
0.01	0.999	0.00897
0.02	0.990	0.0178
0.03	0.955	0.0257
0.04	0.850	0.0305
0.05	0.680	0.0305
0.06	0.480	0.0259
0.07	0.300	0.0188

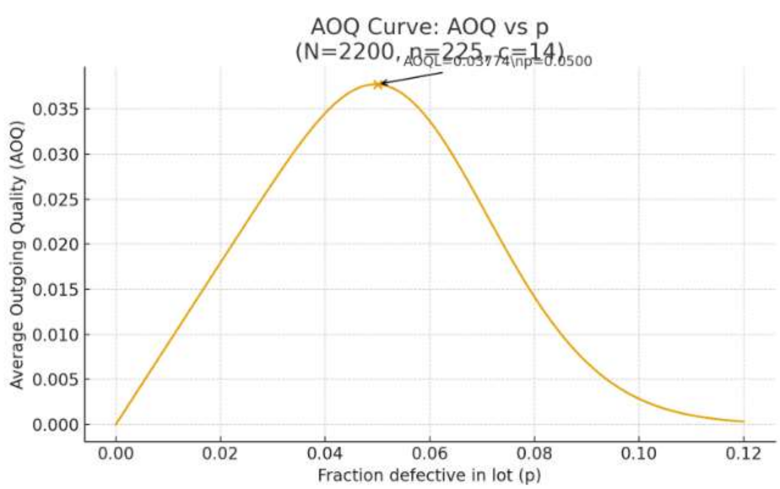
4. AOQL (Average Outgoing Quality Limit)

AOQL = maximum value of AOQ curve.

From above table:

Maximum AOQ ≈ 0.0305 (i.e., 3.05% defective)

This occurs around p = 0.04 – 0.05.



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